

KANGDA WEI

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RESEARCH INTEREST

Natural Language Processing, Large Language Model, LLM Reasoning, Post-Training, Multi-Modal, Agentic Systems

EDUCATION

Texas A&M University	College Station, TX
<i>Doctor of Philosophy in Computer Science. Advisor: Ruihong Huang.</i>	Aug 2023 - Dec 2027 (Expected)
University of North Carolina at Chapel-Hill	Chapel-Hill, NC
<i>Master of Science in Computer Science. Advisor: Shashank Srivastava.</i>	Aug 2022 - May 2023
University of North Carolina at Chapel-Hill	Chapel-Hill, NC
<i>B.S. in Computer Science, B.S. in Statistics and Operational Research, GPA: 3.812/4.0</i>	Aug 2019 - May 2022

INDUSTRY EXPERIENCE

Applied Scientist Intern	Santa Clara, CA
<i>Amazon CS Delphi</i>	
<i>Self-Evolving, Mentor: Hao Chen, Manager: Gokce Keskin</i>	May 2026 - Aug 2026
Research Scientist Intern	Sunnyvale, CA
<i>Bosch Center for Artificial Intelligence (BCAI)</i>	
<i>Multi-Agent Video Understanding, Mentor: Zhengyu Zhou</i>	May 2024 - Aug 2024

RESEARCH EXPERIENCE

Accelerating Mathematical Reasoning Model Training via Diversity-Aware Reward Reweighting

- Developed MMR-GRPO, a training-efficient approach that integrates Maximal Marginal Relevance into Group Relative Policy Optimization to accelerate convergence by prioritizing semantically diverse completions and down-weighting redundant solutions.
- Designed a parameter-free adaptive mechanism that automatically balances diversity and reward quality without hyperparameter tuning, enabling robust performance across different training configurations.
- Achieved 47.9% reduction in training steps and 70.2% reduction in wall-clock training time while maintaining comparable peak performance across three model sizes (1.5B, 7B, 8B), three RL methods (GRPO, DR-GRPO, DAPO), and five mathematical reasoning benchmarks (AIME 2024, AMC 2023, MATH-500, Minerva Math, Olympiad-Bench).
- Demonstrated minimal computational overhead (1-5% per-step) through vectorized GPU operations for embedding extraction and similarity computation, making the approach practical for resource-constrained settings.
- *Status:* Paper submitted to ACL 2026; code and models to be released upon acceptance.

Mitigating Bias and Improving Reasoning in LLMs via Exploratory Thinking and RL

- Proposed a novel data generation framework that encourages exploratory thinking in LLMs by prompting them to reason over structurally identical, morally ambiguous story pairs with male and female protagonists.
- Designed a guided judgement reconciliation process to uncover and mitigate biased moral reasoning, producing balanced gender-neutral story-judgement pairs.
- Performed SFT and DPO on LLMs, achieving substantial gender bias reduction on WinoBias and GenMO benchmarks without sacrificing performance on MMLU and TruthfulQA.
- *Ongoing:* Extending framework to other forms of social bias (e.g., race, social status)

Multi-Agent Systems with LLMs/VLMs for Nuclear Safety Anlysis and Video Understanding

- Designed and implemented multi-agent systems that leverage LLMs and VLMs for two domains:
 - *Nuclear Safety:* Developed a BFS-based multi-agent framework for automatic nuclear fault tree generation in collaboration with Idaho National Laboratory (INL). Built a large-scale nuclear safety dataset via OCR and expert validation; benchmarked multiple LLMs/VLMs for fault analysis tasks.
 - *Video Understanding:* Proposed an end-to-end multi-agent multi-modal framework for advanced indexing of lecture/presentation-style videos. Combined VLM-based shot detection with multi-agent collaboration to capture rich cross-modal information for scalable downstream use. Conducted systematic evaluations, including intrinsic human evaluation (Amazon Mechanical Turk) and extrinsic QA benchmarks, on both public and enterprise datasets.

PUBLICATIONS

- **Kangda Wei**, Ziqian Bi, Ashish Molakalapalli, Haonan Duan, Xiaowei Zou, Vincent Du, Alex Carter, Ruihong Huang. Predicting LLM Training and Inference Runtime: A Survey of Methods, Datasets, and Open Challenges. *In submission*
- **Kangda Wei**, Ruihong Huang. MMR-GRPO: Accelerating GRPO-Style Training through Diversity-Aware Reward Reweighting. *Findings of the Association for Computational Linguistics (ACL) 2026*
- Cheng Zhang, Rajasekhar Kakarla, **Kangda Wei**, Ruihong Huang. ENG-DRB: PDTB-style Discourse Relation Bank on Engineering Tutorial Video Scripts. *Findings of The Asian Federation of Natural Language Processing and The Association for Computational Linguistics 2025*
- **Kangda Wei**, Hasnat Md Abdullah, Ruihong Huang. Mitigating Gender Bias via Fostering Exploratory Thinking in LLMs. *Findings of the Empirical Methods in Natural Language Processing 2025*.
- Abhilekh Borah, Hasnat Md Abdullah, **Kangda Wei**, Ruihong Huang. CliME: Evaluating Multimodal Climate Discourse on Social Media and the Climate Alignment Quotient (CAQ). *NLP4PosImpact Workshop, Association for Computational Linguistics 2025. Best Paper Award.*
- **Kangda Wei**, Xi Shi, Jonathan Tong, Sai Ramana Reddy, Anandhavelu Natarajan, Rajiv Jain, Aparna Garimella, Ruihong Huang. LegalCore: A Dataset for Event Coreference Resolution in Legal Documents. *Findings of Association for Computational Linguistics 2025*.
- **Kangda Wei**, Zhengyu Zhou, Bingqing Wang, Jun Araki, Lukas Lange, Ruihong Huang, Zhe Feng. Multi-Agent Video Understanding for Advanced Indexing of Lecture Videos. *U.S Patent.*
- Hasnat Md Abdullah, Tian Liu, **Kangda Wei**, Shu Kong, Ruihong Huang. UALBench: The First Comprehensive Unusual Activity Localization Benchmark. *Winter Conference on Applications of Computer Vision (WACV) 2025*.
- **Kangda Wei**, Aayush Guatam, Ruihong Huang. Are LLMs Good Annotators for Discourse-level Event Relation Extraction? *Findings of the Empirical Methods in Natural Language Processing 2024*.
- Jiangshu Du, Yibo Wang, Wenting Zhao, ... **Kangda Wei**, et al. LLMs Assist NLP Researchers: Critique Paper (Meta-)Reviewing. *Proceedings of the Empirical Methods in Natural Language Processing 2024*.
- **Kangda Wei**, Dawn Lawrie, Benjamin Van Durme, Yunmo Chen, Orion Weller. When Do Decompositions Help for Machine Reading? *Proceedings of the Empirical Methods in Natural Language Processing 2023*
- **Kangda Wei**, Sayan Ghosh, Rakesh Menon, and Shashank Srivastava. Leveraging Multiple Teachers for Test-Time Adaptation of Language-Guided Classifiers. *Findings of the Empirical Methods in Natural Language Processing 2023*
- **Kangda Wei**, Sayan Ghosh, and Shashank Srivastava. 2022. Compositional Generalization for Kinship Prediction through Data Augmentation. *Proceedings of the 4th Workshop of Narrative Understanding (WNU2022)*, pages 13–19, Seattle, United States. Association for Computational Linguistics.
- *Songhe Wang, ***Kangda Wei**, Lei Lin, Weizi Li. “Spatial-temporal Analysis of COVID-19’s Impact on Human Mobility: the Case of the United States,” in the *20th and 21st Joint COTA International Conference of Transportation Professionals*. *Co-author: equal contribution

PROFESSIONAL SERVICES

- Reviewer for ACL Rolling Review (ARR) since February 2024 (covering ACL/NAACL/EMNLP/EACL/AACL).
- Reviewer for IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.
- Reviewer for Neural Information Processing Systems (NeurIPS), 2026.

TEACHING SERVICES

- TA COMP211 System Fundamentals, UNC-Chapel Hill, Department of Computer Science, Spring 2023
- COMP431 Internet Services and Protocols, UNC-Chapel Hill, Department of Computer Science, Fall 2022

AWARDS

- Outstanding PhD Research Award (one per year), Texas A&M University CSE Department, 2026
- Texas A&M University Graduate School Travel Grant, 2026
- Texas A&M University CSE Department Travel Grant, 2023, 2024, 2025, 2026

SKILLS

- **Programming Languages / Frameworks:** Python, Java, JavaScript, C, HTML, CSS, MATLAB, R, PyTorch, JAX, Hugging Face Transformers, Accelerate, Linux, Parallel Model Training
- **Software & Tools:** PyCharm, VS Code, Spyder, Jupyter Notebook, Weights & Biases,, Docker, Slurm, Git, Tableau, LaTeX, Anaconda, RStudio
- **LLM Training & Post-training:** SFT, RLHF, DPO, GRPO, Reward Modeling, Prompt Engineering, Synthetic Data Generation
- **Cloud & Infrastructure:** AWS, Microsoft Azure, Amazon Mechanical Turk
- **Language:** Chinese (native), English (proficient)